

1. Exploratory Data Analysis (EDA) Summary

The initial phase of the machine learning workflow involved loading and exploring the California Housing dataset to understand its underlying structure and identify potential data quality issues.

- **Missing Values Check:** The dataset was analyzed for null entries. The data was perfectly clean with zero missing values across all 20,640 records, meaning no imputation steps were required.
- **Feature Distributions:** Histograms were plotted to analyze the distribution of each variable. Features such as Population, AveRooms, and AveBedrms exhibited severe right-skewness. The target variable, MedHouseVal (Median House Value), showed a relatively normal distribution but with a distinct, artificial cap at \$500,000, where all higher values were grouped together.
- **Correlation Analysis:** A correlation matrix heatmap was generated to identify linear relationships. The median income (MedInc) demonstrated the strongest positive correlation with the median house value, making it the most critical predictor. Additionally, multicollinearity was identified between several independent variables, such as AveRooms and AveBedrms.

2. Model Architecture and Training

- **Algorithm Selection:** An Ordinary Least Squares (OLS) LinearRegression model from the scikit-learn library was selected to establish a baseline performance. This model attempts to draw a hyperplane that best fits the linear relationship between the census features and the target house prices.
- **Data Splitting:** To ensure the model could be evaluated fairly on unseen data, the dataset was partitioned using the train_test_split function. The data was divided into an 80% training set and a 20% testing set (test_size=0.2), utilizing a fixed random state (random_state=42) for reproducibility.

3. Evaluation Metrics and Visual Diagnostics

The trained model was evaluated against the 20% holdout test set using three standard regression metrics:

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

The model yielded an MAE of approximately **0.533**. Since the target variable is scaled in hundreds of thousands, this indicates the model's predictions are off by roughly \$53,300 on average.

- **Root Mean Squared Error (RMSE):**

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

The RMSE was approximately **0.746**. The higher value relative to the MAE indicates that the model occasionally makes significant prediction errors, penalizing the score.

- **R-Squared (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

The model achieved an R^2 score of roughly **0.595**, meaning that the linear combination of the given features explains about 59.5% of the variance in the California house prices.

Visual Diagnostics:

- **Actual vs. Predicted Plot:** The scatter plot demonstrated a positive linear correlation between the model's predictions and the true values. However, the \$500,000 threshold cap in the true data formed a horizontal ceiling on the plot, where the model struggled to predict accurately.
- **Residuals Plot:** Plotting the errors (residuals) revealed slight heteroscedasticity, where the variance of the errors increased as the predicted house values increased.

4. Improvement Ideas

To improve the model's predictive power and increase the R^2 score, the following strategies should be considered for future iterations:

1. **Feature Transformation:** Applying logarithmic scaling to heavily skewed features would help normalize the data and potentially improve the linear fit.
2. **Regularization (Ridge/Lasso):** To combat the multicollinearity observed in the EDA, implementing L1 (Lasso) or L2 (Ridge) regularization would help constrain the model's coefficients and prevent overfitting.
3. **Non-Linear Algorithms:** Housing prices are influenced by complex, non-linear geographical factors. Replacing the baseline linear regression with tree-based algorithms—such as Random Forests or Gradient Boosting Regressors (e.g., XGBoost)—would likely capture these complex relationships and significantly boost model accuracy.